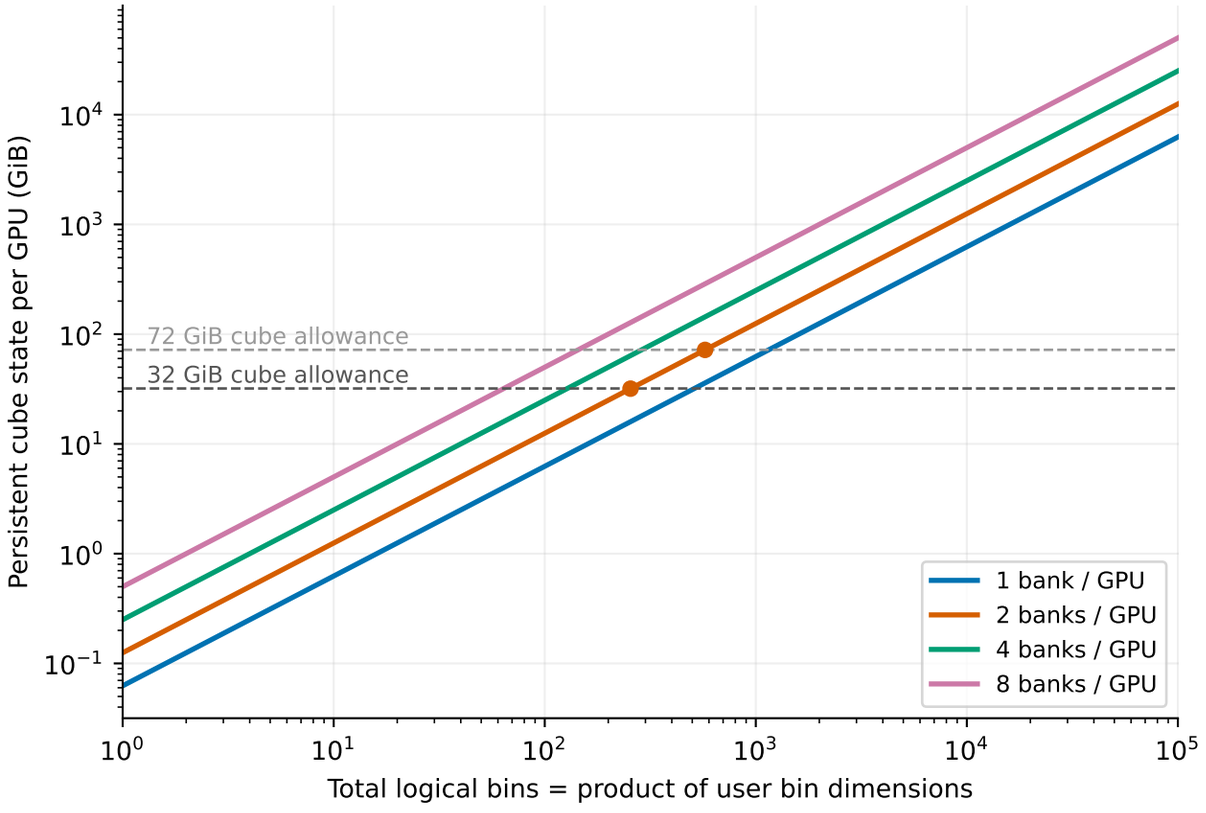
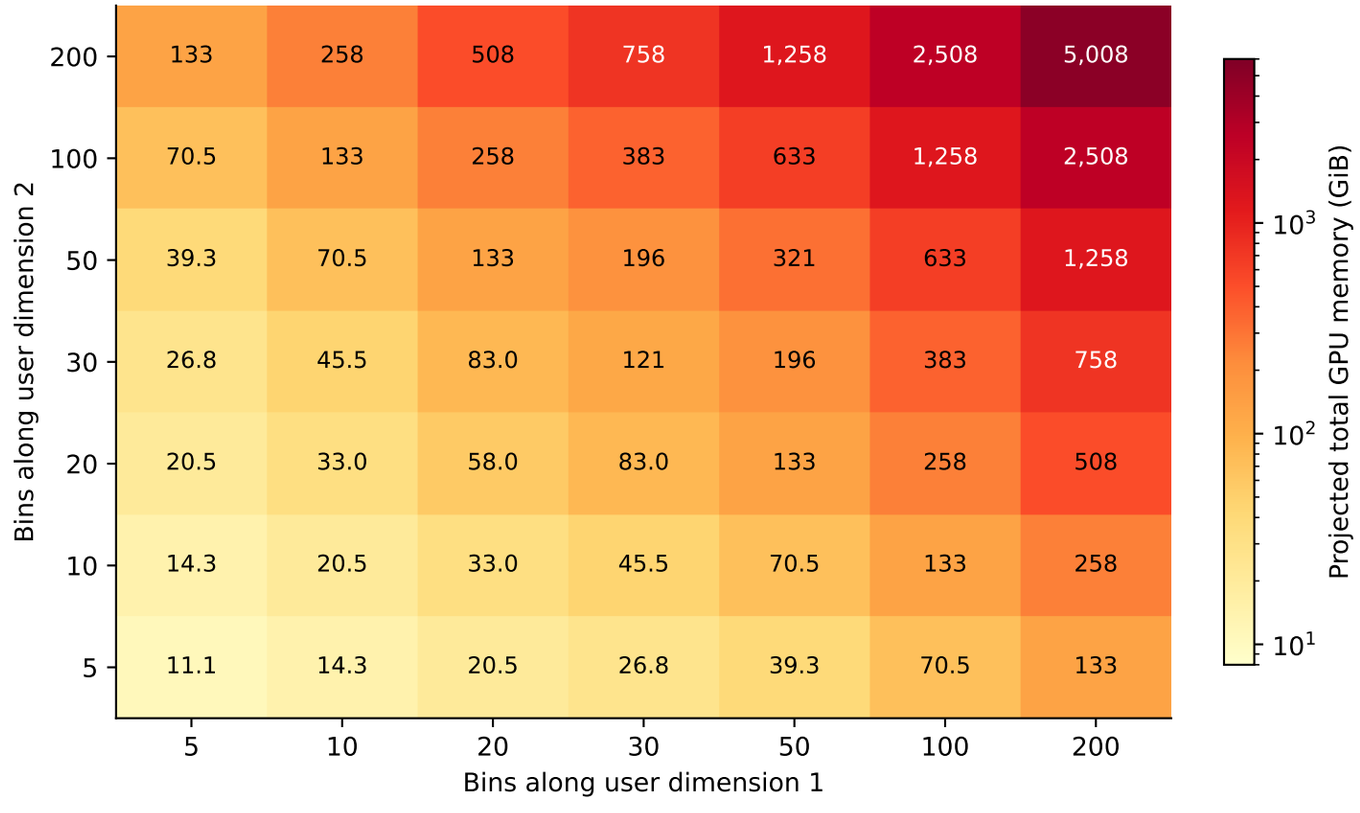
JF hypercube memory: bin dimensions multiply; stream replicas multiply again

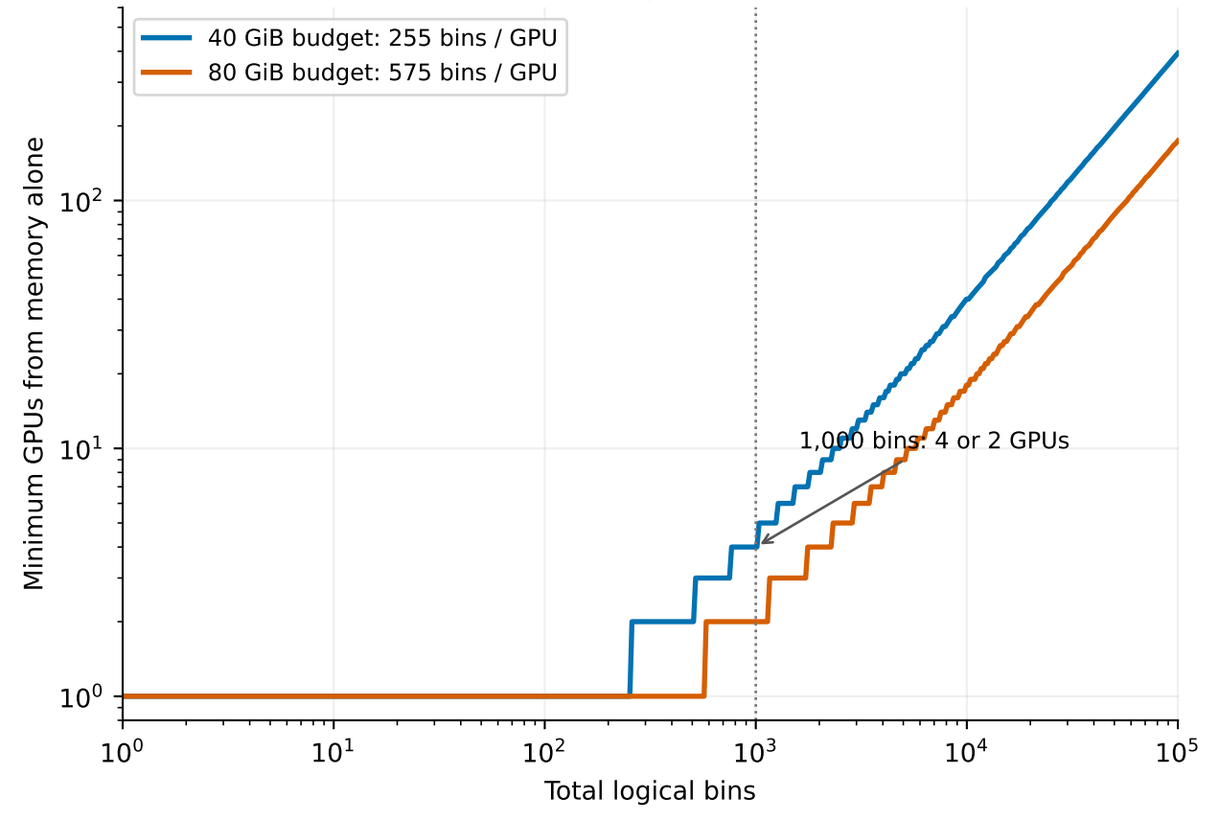
A. Dense replication: float32 sums + scalar counts



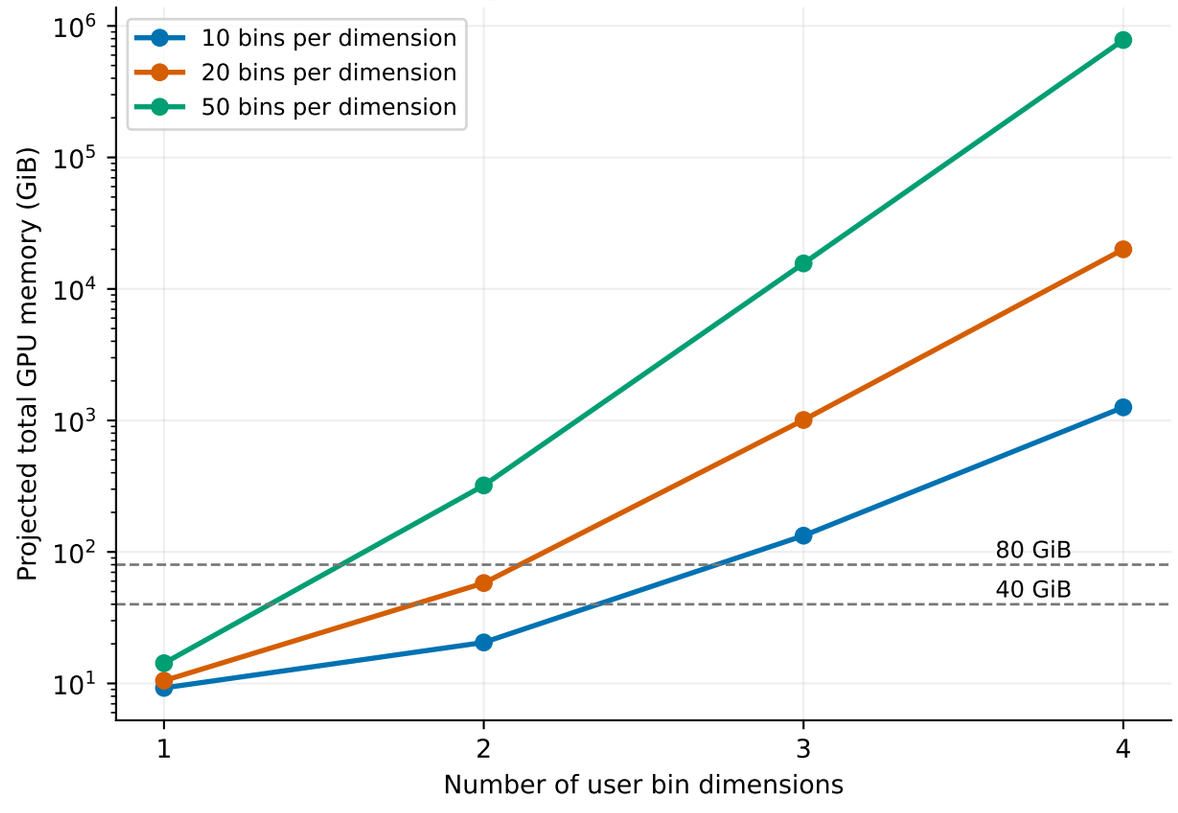
B. Two banks: total GiB including 8 GiB reserve



C. Ideal bin ownership: two banks per GPU



D. Dimension growth: two banks + 8 GiB reserve



Assumptions: 32 MiB uint16 raw → 64 MiB float32 sum/bin; 8 GiB reserve is illustrative.
Sharding requires routing/ownership changes. Counts included; no throughput or actual-device OOM prediction.